

years or a billion years. Resurrection: as you know, Christians believe in resurrection in the flesh, which completely bypasses the need for an immaterial vehicle. So even if one is interested in those questions (and they are not my concern in this paper, although I am concerned to speak to people who have those concerns), even then one doesn't need an immaterial brain or soul-stuff.

So if I am right, and the question of matter or soul-stuff is really irrelevant to any question of philosophical or religious significance, why so much attention to it, why so much heat? The crux of the matter seems to be that both the Diderots of this world and the Descartes of this world have agreed that if we are matter, then there is a physical explanation for how we behave, disappointing or exciting. I think the traditional dualist says '*wouldn't it be terrible if we turned out to be just matter, for then there is a physical explanation for everything we do*'. And the traditional materialist says '*if we are just matter, then there is a physical explanation for everything we do. Isn't that exciting!*' (It is like the distinction between the optimist and the pessimist: an optimist is a person who says 'this is the best of all possible worlds'; and a pessimist is a person who says 'you're right'.)[†]

I think they are both wrong. I think Diderot and Descartes were both wrong in assuming that if we are matter, or our souls are material, then there is a physical explanation for our behavior.

Let me try to illustrate what I mean by a very simple analogy. Suppose we have a very simple physical system – a board in which there are two holes, a circle one inch in diameter and a square one inch high, and a cubical peg one-sixteenth of an inch less than one inch high. We have the following very simple fact to explain: *the peg passes through the square hole, and it does not pass through the round hole.*

In explanation of this, one might attempt the following. One might say that the peg is, after all, a cloud or, better, a rigid lattice of atoms. One might even attempt to give a description of that lattice, compute its electrical potential energy, worry about why it does not collapse, produce some quantum mechanics to explain why it is stable, etc. The board is also a lattice of atoms. I will call the peg 'system *A*', and the holes 'region 1' and 'region 2'. One could compute all possible trajectories of system *A* (there are, by the way, very serious questions about these computations, their effectiveness, feasibility, and so on, but let us assume this), and perhaps one could deduce from just the laws of particle mechanics or quantum electrodynamics that system *A* never passes through region 1, but that there is at least one trajectory which enables

[†] Joke Credit: Joseph Weizenbaum.

it to pass through region 2. Is this an explanation of the fact that the peg passes through the square hole and not the round hole?

Very often we are told that if something is made of matter, its behavior must have a physical explanation. And the argument is that if it is made of matter (and we make a lot of assumptions), then there should be a deduction of its behavior from its material structure. *What makes you call this deduction an explanation?*

On the other hand, if you are not 'hipped' on the idea that *the* explanation must be at the level of the ultimate constituents, and that in fact the explanation might have the property that *the ultimate constituents don't matter*, that *only the higher level structure matters*, then there is a very simple explanation here. The explanation is that the board is rigid, the peg is rigid, and as a matter of geometrical fact, the round hole is smaller than the peg, the square hole is bigger than the cross-section of the peg. The peg passes through the hole that is large enough to take its cross-section, and does not pass through the hole that is too small to take its cross-section. That is a correct explanation whether the peg consists of molecules, or continuous rigid substance, or whatever. (If one wanted to amplify the explanation, one might point out the geometrical fact that a square one inch high is bigger than a circle one inch across.)

Now, one can say that in this explanation certain *relevant structural features of the situation* are brought out. The geometrical features are brought out. It is *relevant* that a square one inch high is bigger than a circle one inch around. And the relationship between the size and shape of the peg and the size and shape of the holes is *relevant*. It is *relevant* that both the board and the peg are *rigid* under transportation. And nothing else is relevant. The same explanation will go in any world (whatever the microstructure) in which those *higher level structural features* are present. In that sense *this explanation is autonomous*.

People have argued that I am wrong to say that the microstructural deduction is not an explanation. I think that in terms of the *purposes for which we use the notion of explanation*, it is not an explanation. If you want to, let us say that the deduction *is* an explanation, it is just a terrible explanation, and why look for terrible explanations when good ones are available?

Goodness is not a subjective matter. Even if one agrees with the positivists who saddled us with the notion of explanation as deduction from laws, one of the things we do in science is to look for laws. Explanation is superior not just subjectively, but *methodologically*, in terms of facilitating the aims of scientific inquiry, if it brings out relevant laws. An explanation is superior if it is more general.

Just taking those two features, and there are many many more one could think of, compare the explanation at the higher level of this phenomenon with the atomic explanation. The explanation at the higher level brings out the relevant geometrical relationships. The lower level explanation conceals those laws. Also notice that the higher level explanation applies to a much more interesting class of systems (of course that has to do with what we are interested in).

The fact is that we are much more interested in generalizing to other structures which are rigid and have various geometrical relations, than we are in generalizing to *the next peg that has exactly this molecular structure*, for the very good reason that there is not going to be a next peg that has exactly this molecular structure. So in terms of real life disciplines, real life ways of slicing up scientific problems, the higher level explanation is far more general, which is why it is *explanatory*.

We were only able to deduce a statement which is lawful at the *higher* level, that the peg goes through the hole which is larger than the cross-section of the peg. When we try to deduce the possible trajectories of 'system *A*' from statements about the individual atoms, we use premises which are totally accidental – this atom is here, this carbon atom is there, and so forth. And that is one reason that it is very misleading to talk about a reduction of a science like economics to the level of the elementary particles making up the players of the economic game. In fact, their motions – buying this, selling that, arriving at an equilibrium price – these motions cannot be deduced from just the equations of motion. Otherwise they would be *physically necessitated*, not *economically necessitated*, to arrive at an equilibrium price. They play that game because they are particular systems with particular boundary conditions which are totally accidental from the point of view of physics. This means that the derivation of the laws of economics from *just* the laws of physics is *in principle* impossible. The derivation of the laws of economics from the laws of physics and *accidental statements about which particles were where when* by a Laplacian supermind might be in principle possible, but why want it? A few chapters of, e.g. von Neumann, will tell one far more about regularities at the level of economic structure than such a deduction ever could.

The conclusion I want to draw from this is that we do have the kind of autonomy that we are looking for in the mental realm. Whatever our mental functioning may be, there seems to be no serious reason to believe that it is *explainable* by our physics and chemistry. And what we are interested in is not: given that we consist of such and such particles, could someone have predicted that we would have this mental

functioning? because such a prediction is not *explanatory*, however great a feat it may be. What we are interested in is: can we say at this autonomous level that since we have this sort of structure, this sort of program, it follows that we will be able to learn this, we will tend to like that, and so on? These are the problems of mental life – the description of this autonomous level of mental functioning – and that is what is to be discovered.

In previous papers, I have argued for the hypothesis that (1) a whole human being is a Turing machine, and (2) that psychological states of a human being are Turing machine states or disjunctions of Turing machine states. In this section I want to argue that this point of view was essentially wrong, and that I was too much in the grip of the reductionist outlook.

Let me begin with a technical difficulty. A *state* of a Turing machine is described in such a way that a Turing machine can be in exactly one state at a time. Moreover, memory and learning are not represented in the Turing machine model as acquisition of new states, but as acquisition of new information printed on the machine's tape. Thus, if human beings have any states at all which resemble Turing machine states, those states must (1) be states the human can be in at any time, independently of learning and memory; and (2) be *total* instantaneous states of the human being – states which determine, together with learning and memory, what the next state will be, as well as totally specifying the present condition of the human being ('totally' from the standpoint of psychological theory, that means).

These characteristics establish that *no* psychological state in any customary sense can be a Turing machine state. Take a particular kind of pain to be a 'psychological state'. If I *am* a Turing machine, then my present 'state' must determine not only whether or not I am having that particular kind of pain, but also whether or not I am about to say 'three', whether or not I am hearing a shrill whine, etc. So the psychological state in question (the pain) is not the same as my 'state' in the sense of *machine state*, although it is possible (so far) that my machine state *determines* my psychological state. Moreover, *no* psychological theory would pretend that having a pain of a particular kind, being about to say 'three', or hearing a shrill whine, etc., all belong to *one* psychological state, although there could well be a machine state characterized by the fact that I was in it only when simultaneously having that pain, being about to say 'three', hearing a shrill whine, etc. So, even if I am a Turing machine, my machine states are *not* the same as my psychological states. My description *qua* Turing machine (machine table) and my